STAT 385 Homework 4 Fall 2026

Due: Oct. 6, 2026

Background & Overview: Breast cancer is the most common cancer diagnosed in women in the United States after skin cancer. Basal-like breast cancer (BLC) is a distinct, aggressive molecular subtype associated with a poor prognosis. The lack of targeted therapies makes treating BLC particularly challenging.

Many human diseases are tied to genomic factors. Richardson et al. (2006) demonstrated that basal-like human breast cancer is strongly associated with X-chromosomal abnormalities, as well as the overexpression of a small subset of X-chromosomal genes.

In this assignment, we use microarray gene expression data as quantitative feature variables to classify whether a patient is normal, has basal-like cancer (BLC), or has non-basal-like cancer (non-BLC). Microarrays allow researchers to measure the abundance of thousands of mRNA transcripts across multiple biological samples simultaneously. Understanding how transcript abundance varies across conditions provides insight into gene functions and regulatory networks.

Proteins serve as the primary functional units of the body, and protein abundance directly correlates with disease phenotypes. Because directly quantifying protein levels is experimentally challenging, microarrays measure mRNA abundance, which correlates positively with protein synthesis, as a surrogate.

Our primary goal is to utilize gene expression data to accurately classify BLC, non-BLC breast cancer and normal types. We will construct, evaluate, and compare four classification models: **Multinomial Logistic Regression**, **Linear Discriminant Analysis (LDA)**, **Naïve Bayes**, and **K-Nearest Neighbors (K-NN)**.

The image below illustrates the biological mechanism linking gene expression microarrays to disease states:

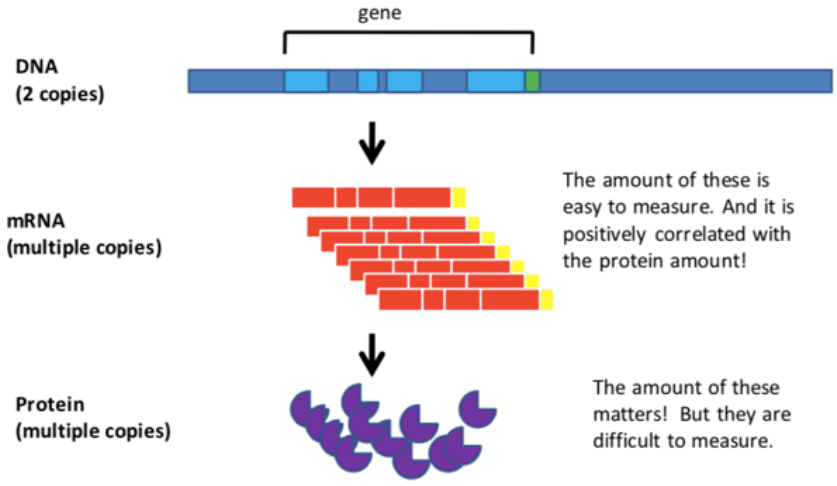


Figure 1: Biological basis for the relationship between mRNA expression and disease states.

Dataset Setup & Instructions:

The dataset `ChromoXmicroarray.txt` is available on UIC Canvas. It contains 1,438 rows and 53 columns, where each row corresponds to a unique gene:

- **Column 1:** Gene ID

- **Column 2:** Gene Symbol/Identifier
- **Columns 3–49:** Gene expression measurements across 47 individual samples
- **Columns 50–53:** Chromosome and physical location details

For this assignment, only the gene expression data in **Columns 3 through 49** will be used. Run the following code snippet in R to load the dataset and configure the response class labels:

```
# Set working directory
setwd("...")

# Load microarray dataset
chromoX <- read.table(file = "ChromoXmicroarray.txt", header = TRUE)

# Construct response labels:
# 1 = Normal (13 samples total)
# 2 = Non-BLC patients (14 samples total)
# 3 = BLC patients (20 samples total)
BLClabels <- c(rep(1, 13), rep(2, 14), rep(3, 20))
```

Questions

Q1. Data Splitting and Reproducibility

Randomly divide the 47 samples into a training set ($n_1 = 35$) and a test set ($n_2 = 12$). To ensure exact reproducibility, execute the following commands prior to splitting [5pts]:

```
set.seed(2026)
train.set <- sample(1:47, 35)
```

Q2. Feature Selection via ANOVA and Feature Normalization

Because the full dataset contains 1,438 genes, reduce the feature space by selecting a small, informative subset [8pts]:

(a) Perform a one-way Analysis of Variance (ANOVA) test on each of the 1,438 genes using only the training set samples to test for mean expression differences across the groups.

(b) Identify the p-values for all 1,438 genes and select the five genes corresponding to the $1^{\textbf{st}}$ **to** $5^{\textbf{th}}$ **smallest** p**-values**.

(c) Form a reduced feature matrix of dimension 47×5 using the expression levels of these five selected genes.

(d) Perform Min-Max normalization on all selected features across all 47 samples using:

$$X_{\text{norm}} = \frac{X - \min(X)}{\max(X) - \min(X)}$$

Q3. Model Training

Using the normalized training dataset from Q2, fit the following classifiers to distinguish BLC, non-BLC and normal cases [12pts]:

- Multinomial Logistic Regression
- Linear Discriminant Analysis (LDA)
- Naïve Bayes
- K-Nearest Neighbors (K-NN) for $k \in \{4, 8, 12\}$

Q4. Decision Boundary Equations

(a) Extract the estimated coefficients from the Multinomial Logistic Regression model and write out the explicit linear equations for the decision boundary lines [5pts].

(b) Extract the projection direction vector from the fitted LDA model and write out the equation for the resulting decision boundary line [5pts].

Q5. Visualization of Decision Boundaries

Plot the training data points overlaid with the decision boundaries for LDA, Naïve Bayes, and K-NN ($k = 4, 8, 12$) [20pts]. You may visualize the boundary:

- Choose two feature variables (say X_1 and X_2)
- Fix the remaining variables at typical values (mean, median, etc.)
- Draw the boundary in the X_1 and X_2 plane.

Note that for Naïve Bayes and K-NN, the separation boundaries between classes may not have a simple parametric form.

- Because Naïve Bayes and K-NN produce non-parametric decision surfaces, construct a fine 2D grid over the feature space, predict class labels across the grid points, and use these predictions to render the decision regions and boundary contours.
- Generate distinct decision boundary plots for K-NN using $k = 5$, and $k = 12$.

Q6. Test Set Evaluation & Performance Comparison

Apply the trained Multinomial Logistic Regression, LDA, Naïve Bayes, and K-NN ($k = 4, 8, 12$) models to predict the class labels for the holdout test set defined in Q1 [50pts]. For each classification method:

(a) Construct confusion matrices and report the number of **False Positives (FP)** and **False Negatives (FN)**.

(b) Evaluate and compare the classification performance using:

- Confusion matrix
- Precision, recall, and F1-score

For the false positives, false negatives, precision, recall, and F1-score calculations, compare the performance for the normal group versus the non-normal group (combining BLC and non-BLC samples).